

OrganicHAR

Towards Activity Discovery in Organic Settings for Privacy-
Preserving Sensors

Prasoon Patidar · Riku Arakawa · Ricardo Graça · Rúben Moutinho
Adriano Soares · Ana Vasconcelos · Filippo Talami · Joana Couto da Silva
Inês Silva · Cristina Mendes Santos · Mayank Goel · Yuvraj Agarwal

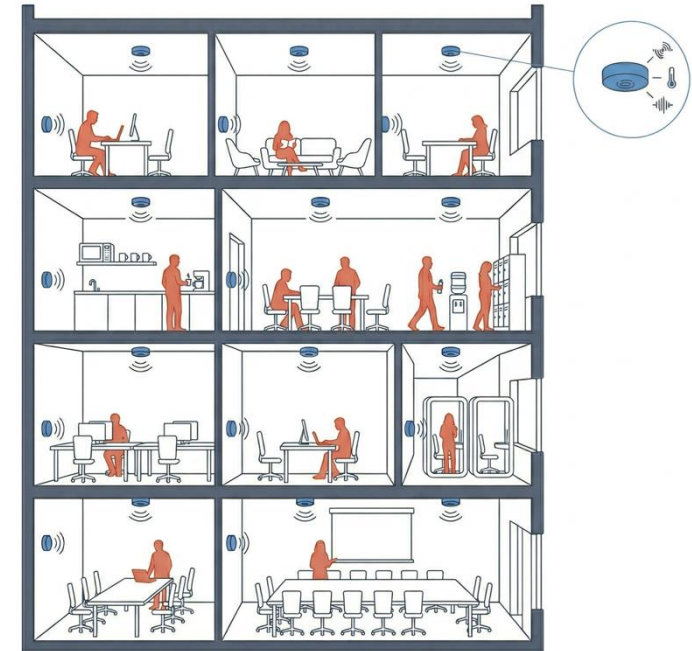
From sensing to understanding



Supporting Safety Monitoring / Task Assistance for Elderly Care



Helping you understand your own productivity patterns



Helping building occupants to answer questions about comfort / occupancy etc.

We need system(s) that observes an environment and builds their own understanding of what's happening.

Deployment considerations at scale



Privacy

"People won't accept 24/7 cloud-connected cameras"



Labeling cost

"Annotating data in every new environment doesn't scale"



Activity variation

"What matters in one home isn't the same as another
and it shifts with time"

Emerging sensing modalities for human activity recognition



DOPPLER RADAR
(MOVEMENT)



LIDAR
(POSITION)



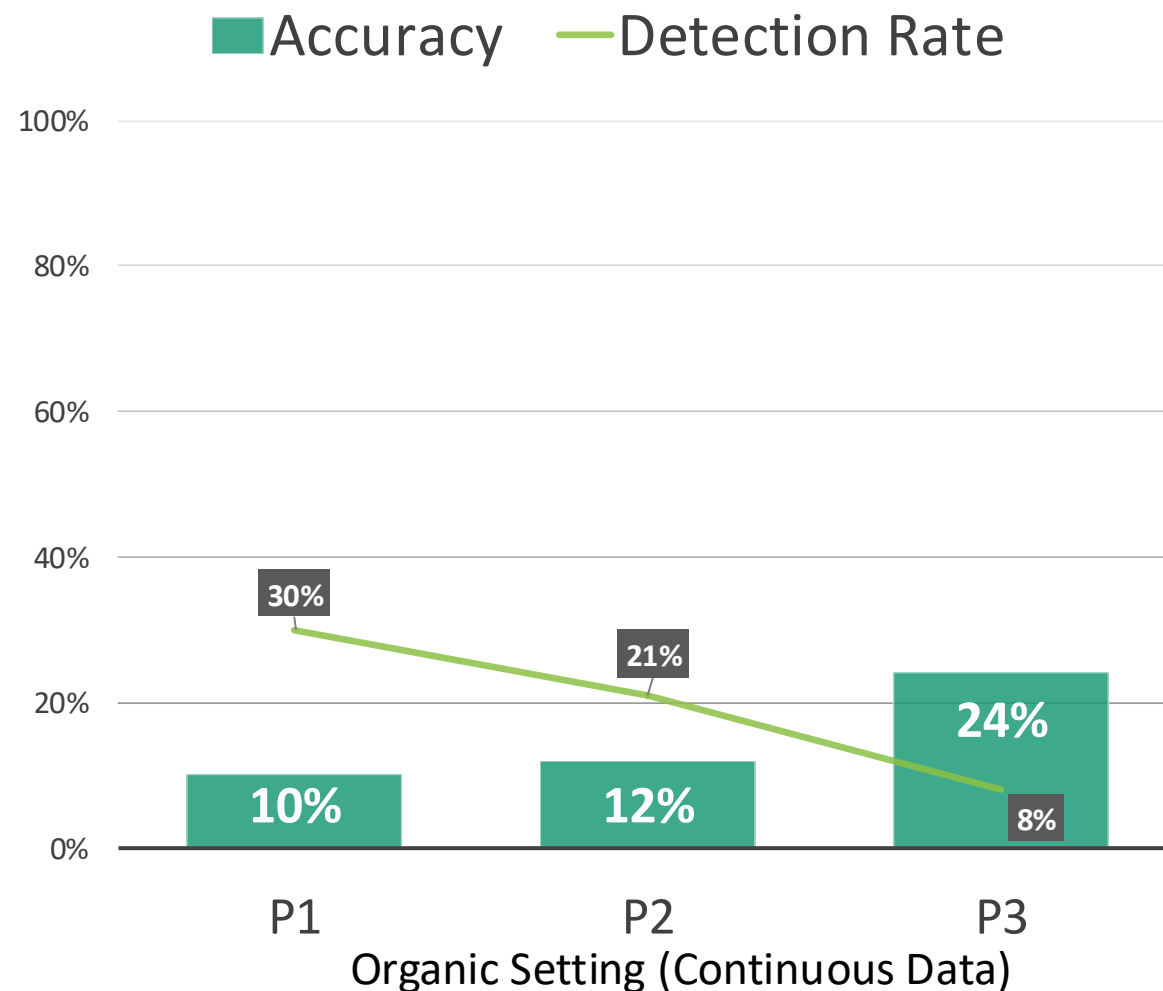
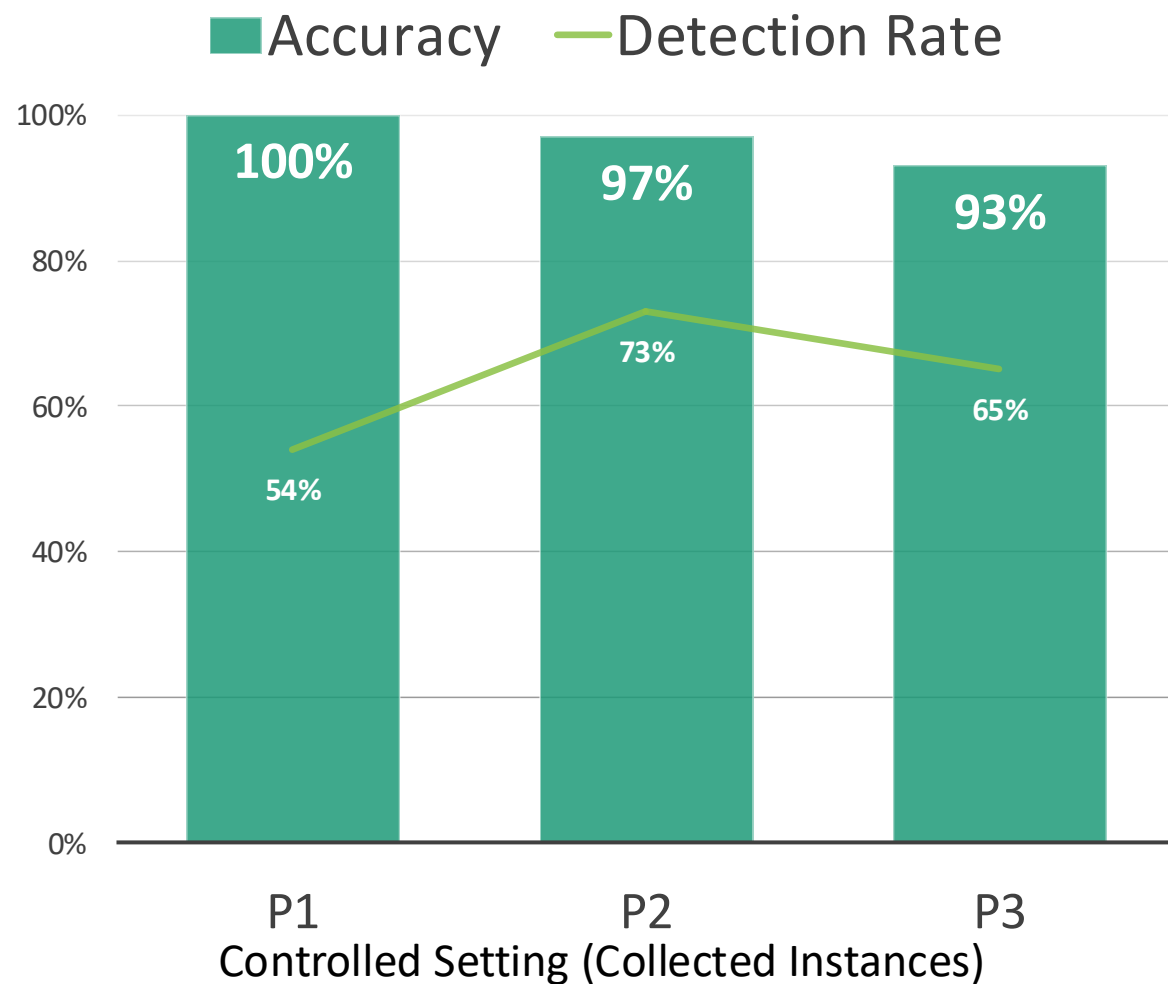
INFRARED
(THERMAL)



However, these modalities need in-situ training data since:

- (a) No existing ML models, activities labels needed
- (b) ML models do not generalize in new placements or locations

HAR Performance in Organic Settings



The challenge of pre-defined activity sets



Same sets of activities are not *meaningful* in new environments.



Same sets of activities are not *detectable* in new environments.



Different applications need different granularity. (coarse-grained vs fine-grained)

Idea: Instead of pre-trained A/V Ensemble with pre-defined activities, use Vision Language Models to get activity labels?

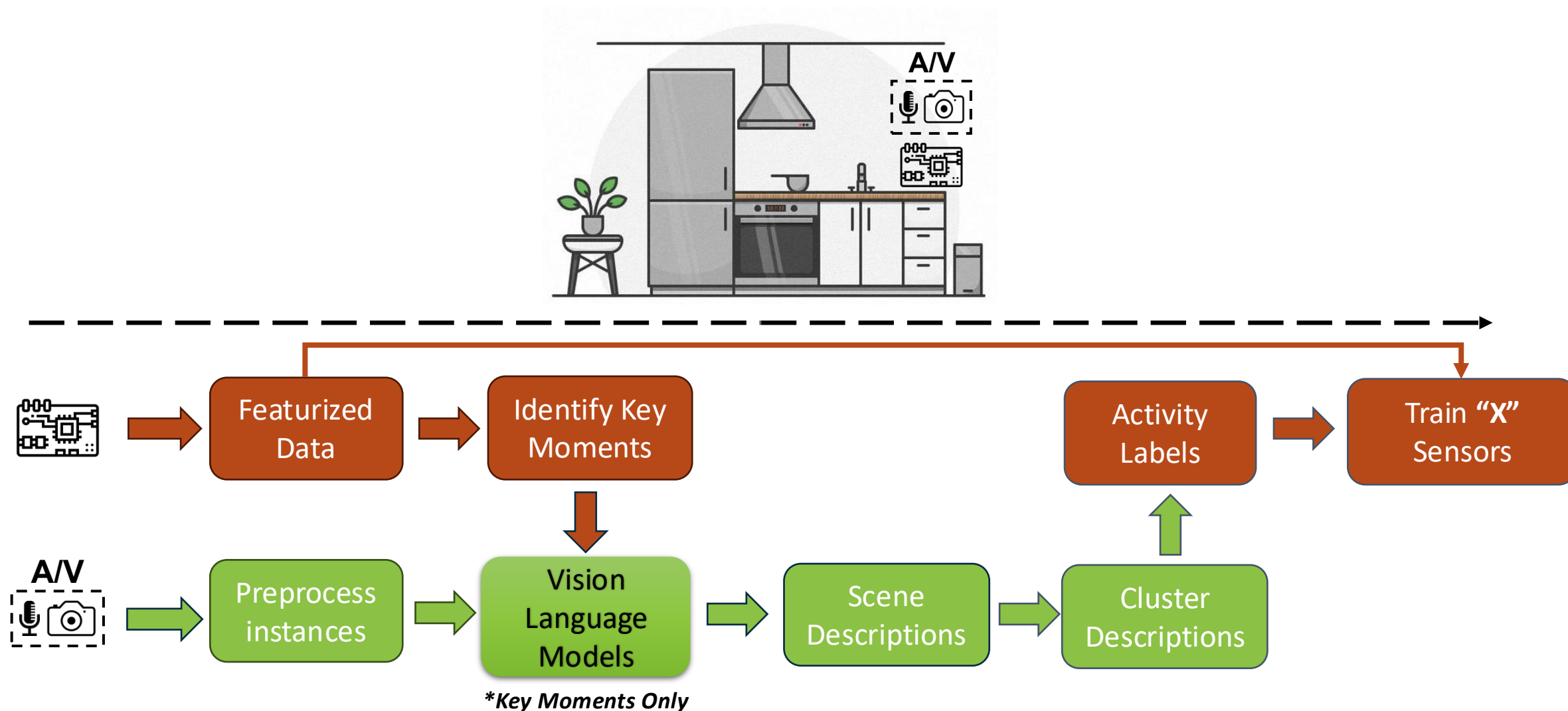
Challenge 1: VLMs are prohibitively expensive. Can we optimize for “when to use VLMs”?

Challenge 2: How to convert VLM outputs (scene descriptions) to clean labels?



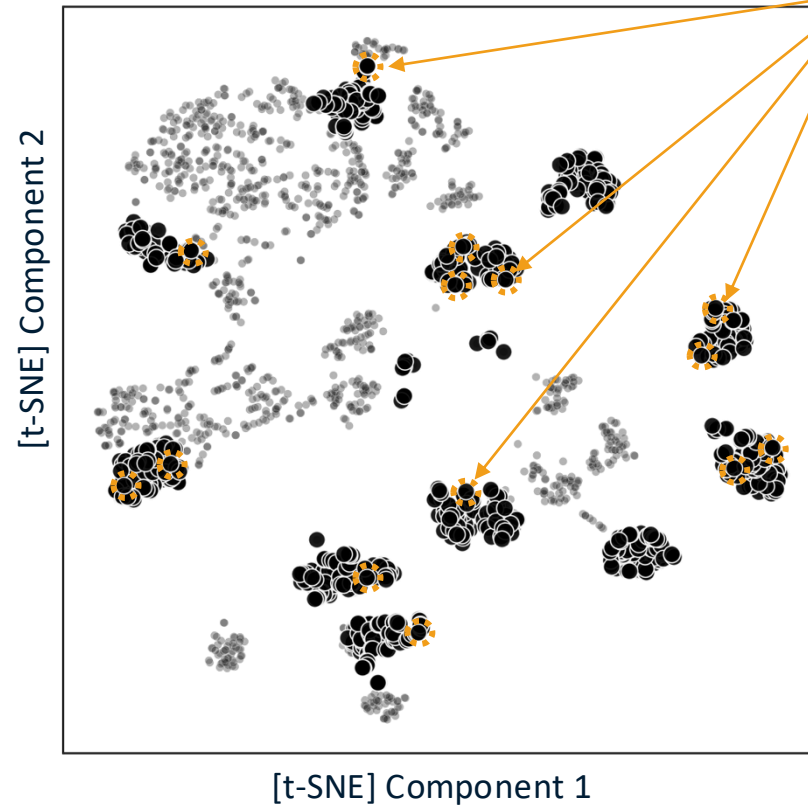
*Using “X” sensors to identify key moments and use (A/V) to **discover** activity labels in new surroundings.*

OrganicHAR: Discovering “meaningful” & “detectable” activity labels

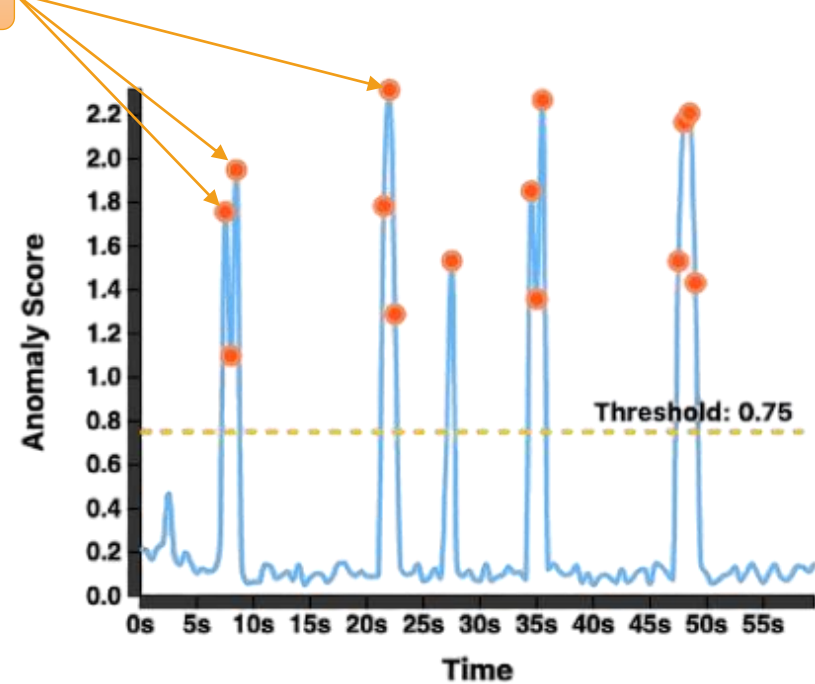


Detecting key moment to collect labels using “X” sensors

Spatial Pattern Discovery
HDBSCAN clustering in feature space



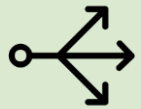
Temporal Transition Detection
Online GMM-based anomaly detection



- Works in high-dimensional sensor feature space
- Finds patterns without predefined cluster count

- Continuous adaptation with forgetting factor $\alpha=0.1$
- Anomaly score based on likelihood under current GMM

Generating labels from VLMs is non-trivial



Non-deterministic outputs for same request



Hallucinating information that is not there or needed



Unstructured natural language responses

From VLM descriptions to activity labels

VLM Descriptions

Noisy, speculative, inconsistent

"washing dish at sink in the morning"
"scrubbing plate with sponge vigorously"
"rinsing cup under faucet with warm water"
"washing hands with soap before cooking"
"filling water glass at sink for hydration"
"cleaning kitchenware after dinner"
"using coffee machine to make espresso"
"pouring cereal into bowl for breakfast"
"making coffee with mug at counter"

Location Consolidation

LLM merges location references → zones

"washing dish at sink"
"scrubbing plate with sponge"
"rinsing cup under faucet"
"washing hands with soap"
"filling water glass at sink"
"cleaning kitchenware"
"using coffee machine"
"pouring cereal into bowl"
"making coffee with mug"

Global
Clustering
(LLMs)

● Sink area ● Counter area

Activity Consolidation

Weighted similarity: action + object + purpose

Sink area

Washing dishes

"washing dish...", "scrubbing plate...",
"rinsing cup...", "cleaning kitchenware"

Washing hands

Kept separate: different purpose

Filling water

"filling water glass at sink"

Counter area

Preparing coffee

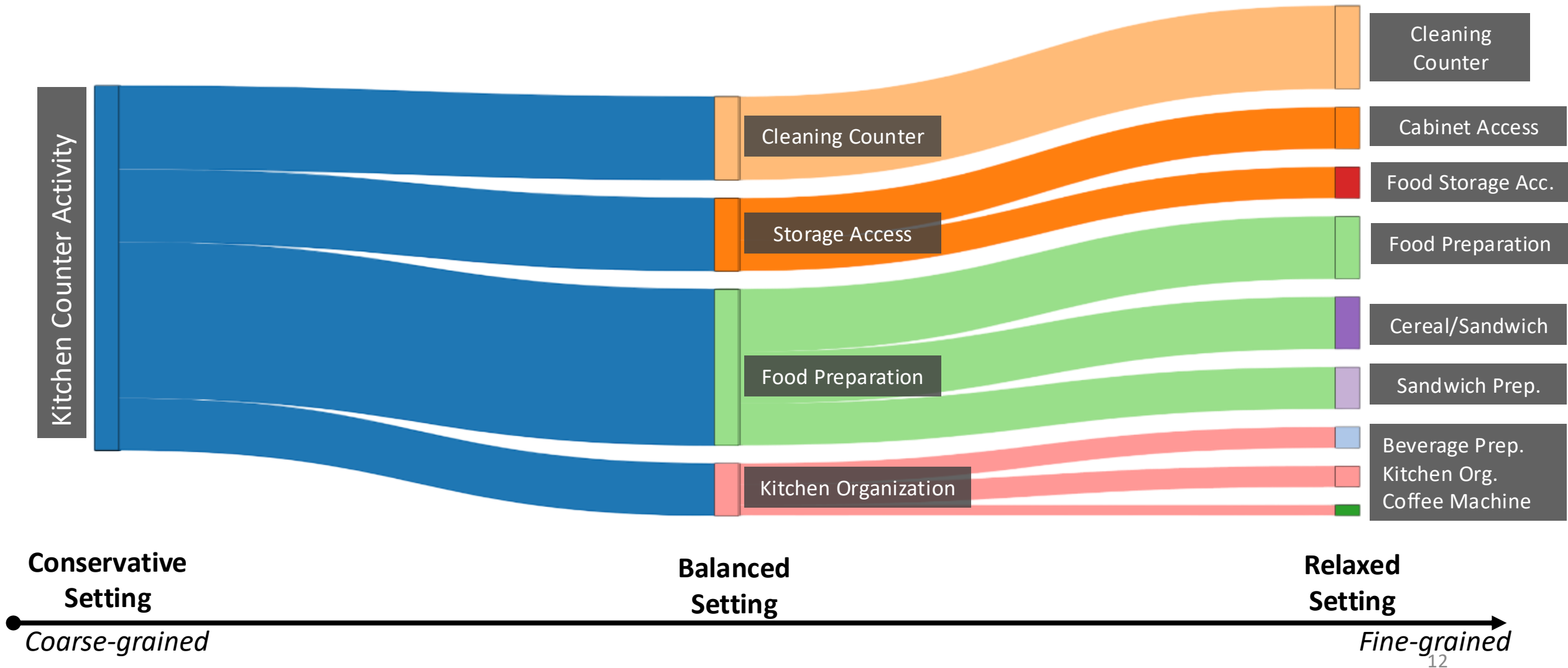
"using coffee machine...", "making coffee..."

Breakfast prep "pouring cereal..."

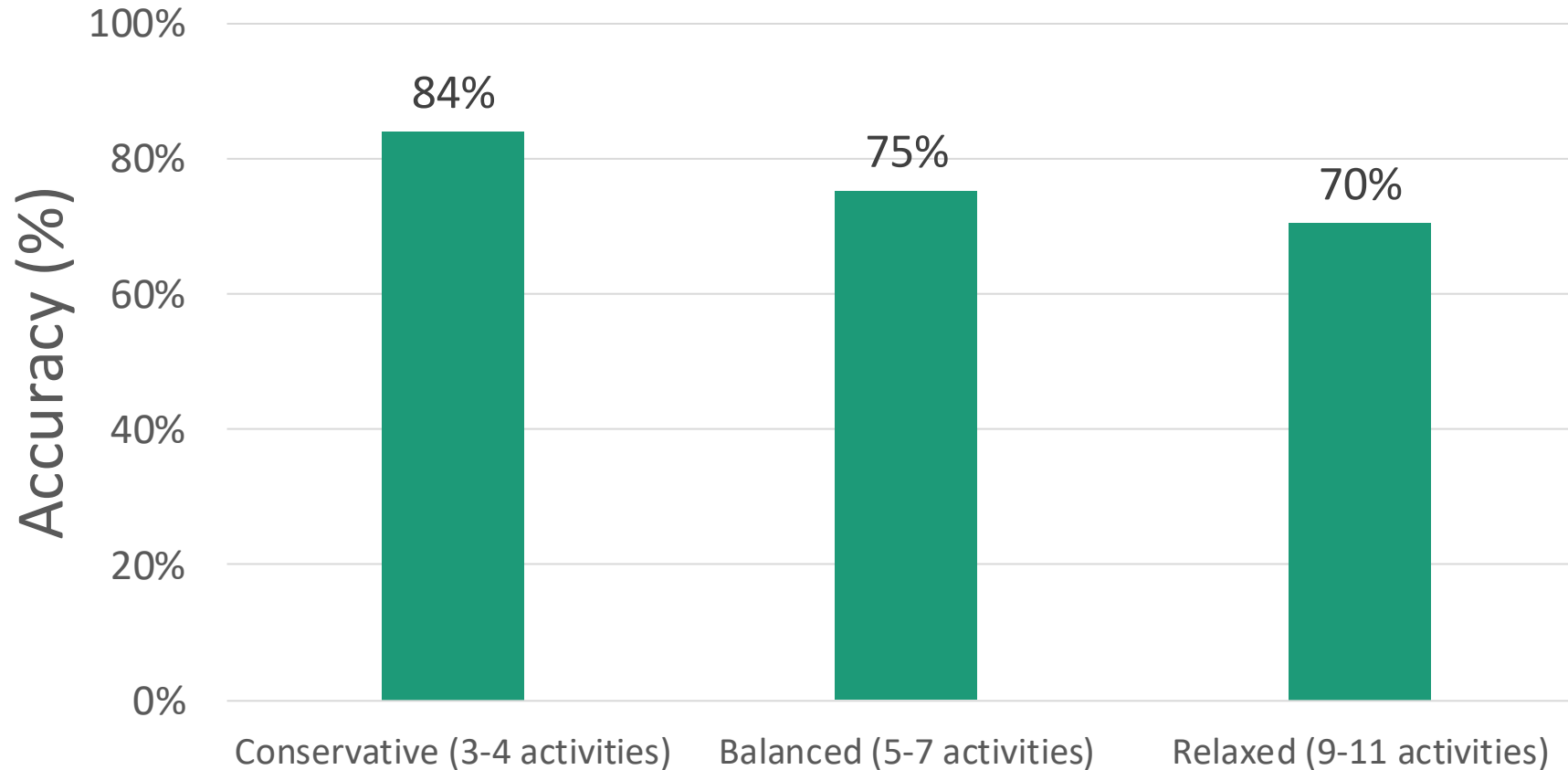
Semantic
Matching

Semantic
Matching

Choosing the right level of granularity for activity labels

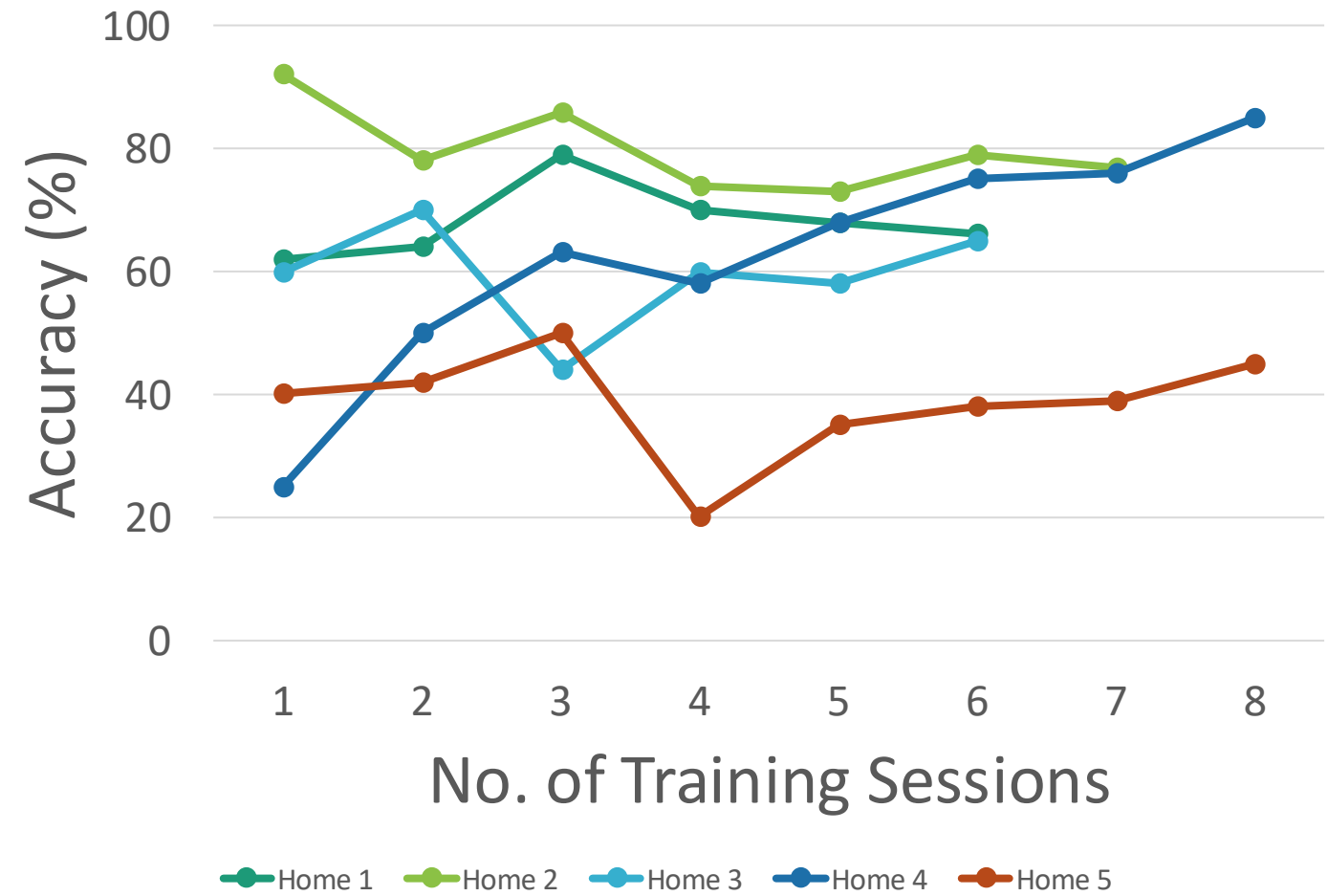


HAR Performance in Organic Setting



(a) Decrease in performance from coarse to fine grained activities.
Recall, in VAX, 84% on pre-defined activity labels.

Long Term Deployment



Home 1



Home 2



Home 3



Home 4



Home 5

Real World Impact: Enabling support for dementia care



Track actions

Records daily activities discreetly, understanding routines and behaviors.



Detect Changes

Recognizes shifts or interruptions in tasks



Offer Guidance

Uses personalized cues like sound, images, and vibration to assist in completing tasks

Safety Monitoring (24/7)

- Not entering kitchen
- Faucet on
- Heat source on
- Fridge left open
- Not eating / Over eating
- ...

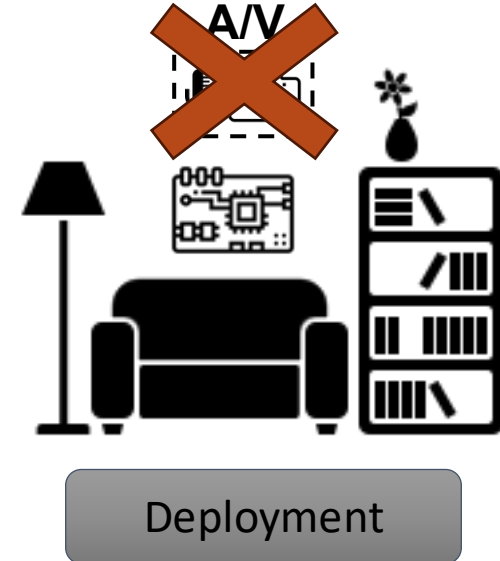
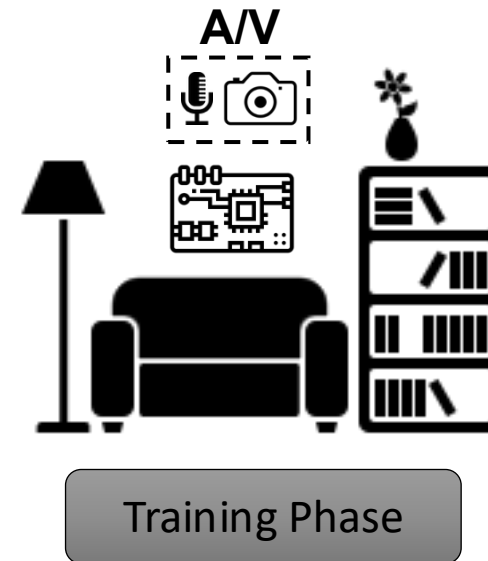
Support (On demand)

- Sandwich
- Hot Beverage (coffee; tea)
- Reheating (microwaving)
- ...

A personalized solution designed to sustain the independence of people with dementia in daily routines.

Conclusion

An end-to-end system to utilize opportunistic labeling with VLMs to discover activities in new environments.



OrganicHAR is open source! Visit the link on QR Code for more details.

<https://github.com/synergylabs/OrganicHAR>

